Milestone 1 — Project Proposal

MuseumCaption: Style-Conditioned Image Captioning using CLIP + GPT-2

Neural Network and Deep Learning — Project 2

Group 5

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Github Repository:
`https://github.com/SimranaSinha/Museum-Canvas-Caption-CLIP-GPT2`

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1. Project Proposal

MuseumCaption is a style-conditioned image captioning system that generates captions for images in three distinct voices: *Art Critic*, *Casual Visitor*, and *Children's Guide*. The system pairs a CLIP image encoder with a GPT-2 language decoder, using style prefix prompts to control the tone of generated captions without modifying the model architecture.

Research Question:

> *Can a CLIP + GPT-2 pipeline generate stylistically distinct captions for the same image using style prefix conditioning — and how does caption quality vary across prompt-only (M1), embedding injection (M2), and fine-tuning approaches (M3)?*

2. Dataset

Property	Detail
Source	HuggingFace: `clip-benchmark/wds_flickr30k`, split = `test`
Full Size	31,000 images with reference captions
M1 Subset	1,000 image–caption pairs
Caption Filter	Captions with fewer than 5 words are discarded
Image Format	PIL RGB, streamed directly from HuggingFace
Cache	Saved to Google Drive as `flickr30k_1k.pkl`

Table 1: Dataset — Flickr30k

3. Model

Component	Detail
Image Encoder	CLIP ViT-B/32 (`openai/clip-vit-base-patch32`) — outputs 512-dim L2-normalized embedding
Language Model	GPT-2 base, 117M parameters (`gpt2`) — no fine-tuning at M1
Style Control	Natural language prompt prefix prepended at inference time
Decoding	Nucleus sampling: top-p=0.92, temperature=0.85, repetition_penalty=1.2, max_new_tokens=60
M1 Status	Prompt-only conditioning — CLIP embedding extracted but not yet injected into GPT-2
Next Step (M2)	Inject CLIP embedding as soft prefix into GPT-2 token sequence

Table 2: Model configuration

Style Prompt Prefixes:

Style	Prompt Prefix
Art Critic	`"An art critic observes this painting:"`
Casual Visitor	`"A museum visitor says about this painting:"`
Children's Guide	`"A tour guide explains this painting to children:"`

Table 3: Style prompt prefixes used in `generate_caption()`

4. Team Roles

Member	Role	Milestone 1 Contribution
Aditya Kumar	Training & Evaluation	Evaluation metric setup: BLEU, METEOR, CIDEr, ROUGE-L
Meghana Rao	Demo & Report	M1 report, README, GitHub repo setup
Simran Abhay Sinha	Data & Preprocessing	Flickr30k pipeline, caption filtering, embedding extraction and Drive caching
Sushmitha S.	Model Architecture	CLIP + GPT-2 integration, style prompts, `generate_caption()` helper

Table 4: Team roles and contributions

5. Sample Caption Test Runs

The figure below shows 5 baseline caption test runs on Flickr30k images using GPT-2 with casual style prompt prefix (no fine-tuning, no embedding injection).

Figure 1: Baseline Generated Captions

#	Image Content	GPT-2 Generated Caption
1	Man wearing a knitted hat with a beer logo	"I thought I was going to have a bit of fun with it..."
2	Black and white dog running in a park	"I was in the park, and there were two of them..."
3	Two people practicing taekwondo on a mat	"You can't look at the whole piece in one sitting..."
4	Group of people in ski gear standing in snow	"It's been sitting there for a long time. We've ne..."
5	Workers on a rooftop	"This is the best work in history. I'd never seen..."

Table 5: 5 baseline caption test runs (casual style, prompt-only, pre-fine-tuning)

Observation: GPT-2 generates fluent text in the casual register but captions are not grounded in image content since no CLIP embedding is injected yet. This motivates the embedding injection work in Milestone 2.